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Saveetha Institute of Medical And Technical Sciences
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ITA0301 Mobile Computing

CO3 AT2 TECHNICAL PRESENTATION

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CDMA Capacity – Problem Statement

Deriving Maximum Simultaneous Users in an Interference-Limited CDMA System

Problem Statement

- A CDMA system has a processing gain (PG) of 256 and requires a linear SIR of 6.
- Derive the maximum number of simultaneous users (N) the system can support.
- Use the relation $SIR = PG / (N - 1)$ for the derivation.
- CDMA (Code Division Multiple Access)
- **Concept**
- All users transmit simultaneously over the same frequency band using unique spreading codes.
- Capacity is limited by mutual interference between users, not by fixed channel slots.
- **Calculation**
- Given: Processing Gain (PG) = 256, Required SIR = 6
- $SIR = PG / (N - 1)$
- Rearranging to solve for N gives the number of users the system supports.



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CDMA Capacity – Step-by-Step Derivation

- **Algebraic Rearrangement**

Start with the relation $SIR = PG / (N - 1)$.

Substitute the known values $PG = 256$ and $SIR = 6$.

This gives the equation $6 = 256 / (N - 1)$.

- **Solving for N**

- Step 1 – Solve for $(N - 1)$:

$$N - 1 = 256 / 6 = 42.67$$

Step 2 – Solve for N: $N = 42.67 + 1 = 43.67 \approx 43$

- **Maximum Simultaneous Users, $N \approx 43$**



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Why Is CDMA Capacity "Soft" Rather Than Hard-Limited?

- FDMA/TDMA assign a fixed number of orthogonal channels or time slots — capacity is a hard ceiling set by spectrum partitioning.
- In CDMA, all **N users** share the same **frequency band** using unique **spreading codes**.
- Capacity vs. interference trade-off:

Interference grows with N: SIR is proportional to $1 / (N - 1)$

- Every added user **raises the interference floor**, which lowers the **SIR for every other active user**.
- As a result, $N \approx 43$ is a **statistical maximum**, not a **fixed hard limit** like an FDMA/TDMA channel count.

What Does "Soft Capacity" Mean Operationally?

- Graceful degradation: exceeding N slightly does not block new users outright — it raises interference and gradually lowers call quality for every active user.
- Cell breathing: a cell's effective coverage and capacity shrink as traffic (interference) rises, and expand again as it falls.



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Conclusion

- **Processing Gain (PG): 256**
- **Required SIR: 6**
- **Maximum simultaneous users, $N \approx 43$** , derived from $(N - 1) = PG / SIR$, rearranged from $SIR = PG / (N - 1)$.
- **Reason:** CDMA capacity is interference-limited: every additional user directly reduces the SIR of all others, so capacity is a "soft," probabilistic ceiling rather than a fixed channel count.
- **"With a processing gain of 256 and a required SIR of 6, rearranging $SIR = PG / (N - 1)$ gives $N - 1 = 256 / 6 = 42.67$, so $N \approx 43$ users. Because this limit comes from interference rather than fixed channels, operators use admission control and accept graceful, quality-based degradation instead of a hard block on new users."**

*Thank
you!*